Forecasting the impact of systematic unvertainties on the dispersion mseasure redshift relation of fast radio bursts

Elyas Farah

Masterarbeit in Physik
angefertigt im Argelander-Institut für Astronomie

vorgelegt der
Mathematisch-Naturwissenschaftlichen Fakultät
der
Rheinischen Friedrich-Wilhelms-Universität
Bonn

MMM 2025

I hereby declare that this thesis was formulated by myself and that no sources or tools other than those cited were used.

Bonn, ...

Date Signature

Gutachterin und Betreuerin: Prof. Dr. Jennifer Schöber
Gutachter: Prof. Dr. Cristriano Poriciani

Contents

1 Introduction **3**

2 Theoretical Background **7**
2.1 The general theory of relativity . . . 7
2.2 The Friedmann-Robertson-Lemaître-Walker metric . . . 10
2.3 The cosmoligcal model . . . 12
2.4 Weak gravitational Lensing . . . 13
2.5 The impact of baryonic feedback on the large-scale structure . . . 13
2.6 Fast radio bursts . . . 13

3 Methodology **15**
3.1 Modelling the electron power spectrum . . . 15
3.1.1 The baryonification correction model . . . 15
3.1.2 The halo model . . . 15
3.2 Simulating the dispersion measure signal . . . 15
3.2.1 the Generator for Large Scale Structure (GLASS) . . . 15
3.2.2 The electron angular power spectra . . . 15
3.2.3 The cosmological covariance of the dispersion measure signal . . . 15
3.3 The systematic uncertainties . . . 15
3.3.1 Redshift uncertainty . . . 16
3.3.2 Selection effects . . . 16
3.3.3 Unlocalised hosts . . . 16
3.3.4 Milkyway contribution . . . 16
3.3.5 Host galaxy contribution . . . 16
3.3.6 The lensing of the Dispersion measure signal . . . 16
3.4 Simulation-based inference . . . 16
3.4.1 Approximate Baysian inference . . . 16
3.4.2 Neural Posterior Estimator . . . 16
3.4.3 The Sequentual Neural Posterio Estimator . . . 16
3.4.4 Flow matching posterior estimator . . . 16
3.4.5 Data Compression . . . 16

4 Results **17**

5 Discussion & Conclusion **19**

A Useful information 21

Bibliography 23

List of Figures 27

List of Tables 29

This is your thesis abstract. It may be in a language that is different from the rest of your thesis.

This is your thesis abstract. It may be in a language that is different from the rest of your thesis.

CHAPTER 1

Introduction

The questions on how did the universe came to be, why is the universe as such, and what is the universe heading towards falls upon cosmology to answer. Cosmology is the study of the universe on large scales in an attempt to answer the latter questions cite maybe Peebles. This pursuit started as a data-starved research cite and transformed into a data-rich endevour as a result of a platitutde of imaging surveys such as the LSST Vera Rubin observatory, Euclid, and Nancy Roman cite. The qualitative questions, with which this paragraph started,< transcribe in the statistical sense to be what is the cosmological model that best describes the universe on throughout length scales and energy scales.

The first attempt to describe the universe as a whole came through the lens of the general theory of relativity where Robertson and Walker cite built on the work of Friedmann and Lemaître to describe geometry of the universe through what is now known as the Friedmann-Lemaître-Robertson-Walker (FLRW) metric. We discuss this metric further in 2.2. This metric was largely motivated by the observations of Edwin Hubble who saw that galaxy further away from the Earth are moving faster than galaxies closer to Earth Hubble, 1929. Figure shows the plot from Hubble's work in 1929.

Further observations of the magnitude of distant galaxies have shown that the universe is expanding cite. The general theory of relativity attributes this expansion to the energy density content of the universe. Therefore, the question of what is driving the expansion of the universe and what are the componenets that contribute to the energy budget of the universe represent a crucial question in cosmology for which multiple cosmological models have been suggested (Di Valentino et al., 2025). In fact, the cosmological model has been motivated by both cosmological and astrophysical observation. Some of these observation are the measurement of galaxy rotation curves by Vera Rubin cite that hinted at an additional matter component that does not interact with light and the discovery of the cosmic microwave background (CMB) radiation that hinted at hot and homogeneous early universe.cite. Furthermore,early observaiton of clusters of galaxies have shown that the spatial distribution of galaxy deviates from randomness. This was shown by Peebles cite. By calculating the two point correlation function, there was clearly a signal in the way that galaxies are distributed which points out to an underlying mechanism in the formation of these galaxies (Totsuji and Kihara, 1969). The development of obsrevational techniques and instrumentation allowed the launch of the Center for Astrophysics Redshift Survey (Davis, Huchra et al., 1982) which showed that galaxy positions are correlated on large scales and that there is a structure in the universe (Davis and Peebles, 1983).

In the early 1990s, cosmology experienced a paradigm shift with the launch of the Cosmic

Background Explorer (COBE) sattelite (Boggess et al., 1992). The COBE mission provided the first detection of primordial CMB anisotropies and further confirmed that the universe is flat (Smoot, 1999). In the context of the FLRW metric, the flatness of the universe requires a total sum of the relative density parameters to be 1 (within error bounds). Therefore, in addition to the matter component (dark and baryons), there should be a third component that can account for the missing energy density. Using supernova data from the Hubble Space Telescope (Bahcall and Spitzer, 1982), A. G. Riess et al. (1998) found a strong evidence for an expanding universe universe with negative acceleration and supported the findings of the COBE mission for the existance of additional density component different from matter which is driving the expansion of the universe. Hereafter, the term 'dark energy was coined to refer to this component (Turner, 2001). Wiht this, the ΛCDM model was born, and the efforts to obtain a precise estimates of the cosmological parameters intensified (Bennett et al., 2003; Pope, 2004).

While early experiements relied on CMB, galaxy number counts, or supernova to infer cosmological parameters, gravitational lensing was in development as a tool to constrain the cosmological parameters. Gravitational lensing is the distortions of light geodesics by gravity when light travels from a distant source travels through the inhomogeneous universe towards an observer on the Earth (P. Schneider, Ehlers and Falco, 1992). On the level, of a galaxy, gravitional lensing effects manifest themselves as a distortion of the shape of the galaxy and a change in the angular position with respect to a particular observer. Kaiser and Squires (1993) showed that these distortions could be used to obtain a map of the integrated mass density along the line of sight. The use of galaxy shape distortions for cosmology was a direct result of the Kairser-Squire inversion formula where Seitz and P. Schneider (1996) realised that the expectation value of galaxy shape is an unbiased estimate of the shear in the weak field limit. In principle, it is possible to relate the power spectrum of galaxy shapes with the matter power spectrum, offering a concrete method of probing cosmology by studying the correlations between galaxy shapes (Bartelmann and P. Schneider, 2001). This new probe of the large scale structure is called cosmic shear and was used by Maoli et al. (2001) to obtain constraints on the cosmological parameters. These constraints were not as tight as the ones obtained by COBE, however, it paved the road towards using cosmic shear to constrain the cosmological model. The advantage of cosmic shear as an integrated quantitity along the line of sight mean that it is sensitive to structure growth in the late universe. Therefore, it represent a very good tool to constrain the equation of state for dark energy (Hu and Tegmark, 1999; Jain and Taylor, 2003).

After that, multiple galaxy surveys used cosmic shear for cosmilogical parameter inference, and numerous statistical tools were developped to deal with the increasing data size (P. Schneider, Van Waerbeke et al., 1998; P. Schneider, Eifler and Krause, 2010). Likewise, with increasing data size, the noise level in a cosmological signal increases and the need to mititgate the impact of the systematic and measreument uncertainties is imperative. In this work, we focus on tackling on of the most crucial systematics, namely baryonic feedback on the large scale structure.

Baryonic feedback are the set of baryonic physics processes that impact the distribution of the matter (Rudd, Zentner and Kravtsov, 2008). These processes include, but not limited to, supernova feedback, active galactic nuclei feedback, and star formation. By analying hydrodynamical simulations, Daalen et al. (2011) realised that baryonic feedback affect the matter distribution in the universe at different scales, meaning that their impact should be modelled and included in a theorey of structure formation. However, the theoretical formalism of structure growth, and therefore the matter power spectrum assumes that structure growth is only driven by gravitional effects, ignoring baryonic feedback (Dodelson and Schmidt, 2026). Modelling baryonic feedback analytically is a cumbersome

task, especially that the physics of galaxy formation and evolution is highly uncertain where different suits of hydrodunamical simulations have a different feedback models, and each one of these model yields a totally different desciption of the matter power spectrum supression (Daalen et al., 2011). The mismodelling of the matter power spectrum tunnels down to the modelled cosmic shear signal and yields biased parameter estimates in a comsmological analysis.

On a conquest for precision cosmology, multiple statistical and modelling methods attempted to mititgate the impact of baryonic feedback on the large scale structure. For example, Krause et al. (2021) imposed scle cuts on measured shear correlation funciton and only considered larger scales where baryonic feedback is minimal. However, the cosmic shear signal derives most of its sensitivity from non-linear scale i.e. small scales (Piccirilli et al., 2025). Therefore, imposing a scale cut lead to loss of information. Other statistical techniques involve the use of principle component analysis to compress the parameter space of baryonic feedback parameters from hydrodynamical simulations and marginlising over these parameters in a cosmological analysis (Eifler et al., 2015). Another statistical method involves introducing an additive term to the covariance matrix called the theoretical error covariance (Moreira et al., 2021; Maraio, Hall and Taylor, 2025) which descibe the variation of different baryonic feedback scenarios among all available hydrodynamical simualtions. Analyses involving all the afformentioned results yields unbiased estimates of the cosmological parameters, however, it comes at a cost of introducing additional scatter due to the existing nuisanc parameters. Finally, A. Schneider and Teyssier (2015) introduced the baryonification correction model, a model that parametrizes the displacement of dark matter particles in the existance of baryonic feedback. This model and its developments (A. Schneider, Kovač et al., 2025) were calibrated on hydrodynamical simulations and were able to emulate baryonic feedback in other hydrodynamical simulations that they were not calibrated on. Having this model, it is possible to obtain a prediciton for any probe of baryon in the universe. In particular, this model was used to constrain baryonic feedback parameters with X-ray and the kinematic Sunyaev-Zel'dovich effect (Kovač et al., 2025) since it is possible to obtain a prediction of gas power spectra form which a simulated signal of the correponding baryon tracer could be obtained. Though these probes are promising, they rely on some assumptions with respect the structure of the galaxy clusters such as having hydrodynamic equilibrium and require many convolution from a prediciotn of the gas power spectrum up to a predicted signal. Luckily, fast radio bursts provide a direct probe of the electron density in the universe through the information that their signal encodes.

Fast radio bursts (FRBs) are pulses of electromagnetic waves of extragalactic origin. First discovered in 2007 (Lorimer et al., 2007), the signal of FRBs gets broadened while travelling through the universe, meaning that that the wave packets of higher frquency will arrive later that wave packets of higher frquency. This difference in arrival time is directly proportional to the integrated electron density along the line of sight where the dispersion measure signal could be easily simulated with minimal assumptions about the relationship of the dispersion measure and the electron number density in the universe. The grand benefit of the dispersion measure signal lies in its constraining power of baryonic feedback parameters since it is a drirect tracer of the baryon distributino in the universe. Even though the source of these bursts is unknown (Petroff, Hessels and Lorimer, 2019), it is still possible to exploit the dispersion measure signal in a cosmological context, given that FRBs are hosted within galaxies whose redshift can be identified.

Simulated likelihood, analyses showed that the cross-correlation of FRBs with other probes of structure yields better constraints on the some cosmological parameters. In particular, Calibrating baryonic feedback with FRBs and weak lesning led to one order of magnitude better constrains on the

baryonic feedback and 50% improvements on constraints of the sum of neutrino masses $\sum m_\nu$ and the baryonic fraction in the unvierse Ω_b (Reischke, Neumann et al., 2023). Furthermore, combining FRBs with galaxy number counts in a 3x2-point analysis provided better constraints on baryonic feedback parameters as compared to galaxy number counts (Sharma et al., 2026). In practice, we can aim to use FRBs alone as a probe to impose tigh constraints on the baryonic feedback parameters and use these constraints as priors in a cosmological analysis involving probes of structure, allowing the information content of LSS data to constrain the cosmological parameters and delegating the task of constraining baryonic feedback to FRBs.

The promising potential of FRBs in constraining baryonic feedback could be hindered by multiple modelling and measurement uncertainties. First, FRBs are observed using radio telescopes with only limited follow-up optical observations that are necessairy to calculate the redshifts of galaxies and by turn quantify the associated mean DM at that redshift. Second, measreuments of these redshifts are highly uncertain especially for sources at high redshifts. Third, the measured disperison measure signal receive three main contribution: a contribution from the Milkyway, a contribution from the host galaxy, and a contribution from the circugalactic medium which is the quantity of comsological interest (Reischke and Hagstotz, 2025). The models for both the Milkyway and the host galaxy contribution are both uncertain, leading to biases in the parameters constraints if not dealth with correctly. Fourth, the theoretical formalism of fast radio bursts assumes that the radio signal travels in a homogeneous universe, ignoring any gravitational lensing effect due to large scale structure. There are contradicting resutls on impact of gravitaional lensing; Alonso (2021) showed that first order contribution from general relativitstic effects to the DM angular power spectrum are subdominant compared to the contribution from the electron column density, meanwhile, Takahashi (2024) showed that the second order contribution due to lensing could lead to 1% diviation in the luminosity distance, resulting underestimation of the the luminosity distance.

Therefore, the purpose of this thesis is to model, forecast, and explore mitigation effects of systematic uncertainties and describe thrir importance with varying sample size. With upcoming radio survey, a plathora of data will be available that an accurate modelling of the dispersion measure signal is crucial to extract meanigful information from the signal (Luke Hoffmann et al., 2024). I scale up previoud work by Konar et al. (2025) by using simulation-based inference (SBI), hereafter to infer the impact of systematic uncertainties on the baryonic parameters constraints using fast radio bursts only. Using SBI in this context has multiple benefits. First, it facilitates integrating different uncertainty effects on the level the dispersion measure signal either on the level of the dispersion measure or the level of the redshift. Second, it retires the necessity to approximate the likelihood of the data given the parameters unlike traditional Baysian analyses. Results from ray tracing in hydrodynamical simualtions (Konietzka et al., 2025) suggests that a Gaussian or lognormal approximation to the probability distribution function breaks down for large number of FRBs, and analytic models of the dispersino measure is needed to simulate teh PDF of FRBs. (Torkamani et al., 2026). More details on SBI and the method used in this thesis could be found in 3.4

This thesis is structured as follows: I review the theoretical background in 2 including the general theory of relativity (2.1), the FLRW metric (2.2), the cosmological model (2.3), weak gravitational lensing (2.4), baryonic feedback on the large scale structure (2.5), and basics of fast radio bursts (2.6). I describe the methodology in 3 including modeeling the electron power spectrum (3.1), simulating the dispersion measure signal (3.2), describing the systematic unvertainties (3.3), and explainging simulation based inference (3.4). In 4, I describe the results. Finally, I dicuss the results in 5

CHAPTER 2

Theoretical Background

This chapter is mainly based on 2 textbooks and 2 lectrue notes. The first book is **Modern Cosmology by** Dodelson and Schmidt (2026) and the second is **Spacetime and Geometry: An Introduction to General Relativity** by Carroll (2024). The first lecture notes come from the Advanced Cosmology course given the the Summer semester 2024 by Prof. Dr. Cristiano Poriciani and Jun-Prof. Andrina Nicola, and the second lecture notes come from the Weak Lesning course by prof. Dr. Peter Schneider given in the In this chapeter, I will introduce the basics of the principles of cosmology, weak gravitational lensing, and fast radio bursts (FRBs).

2.1 The general theory of relativity

The general theory of relativity introduces a geometric interpretation of gravity, resting on a set of geometric assumptions and principles evolved from the older classical Newtonian theory of gravity and relativity. The first of these is the **weak equivilance principle** which states that **' the inertial mass and gravitational mass of any object are equal'**. Einstein reformulated this statement to fir the language of the special theory of relativity by stating that **'In small enough regions of spacetime, the laws of physics reduce to those of special relativity; it is impossible to detect the existence of a gravitational field by means of local experiments.'** The consequences of this principle is that freely falling frame of reference cannot recognize that freely falling test particles in the vicinity of the frame are being accelerated (or under the influence of gravity). The consequences of this principle invites the formulation of a geometric theory of gravity where gravity manifests itself as a curverture.

To have a mathematical theory where observation could be interpreted, Einstein used differentiable manifolds to describe the four dimensional spacetime. Differentialble manifolds allows a definitation physical quantities such as position and momentum along with a continuous derivative of these quntities on the manifolds. Having a differetiable quantity allows also to introduce conservation laws which are a direct results of symmetries in a particular problem (Noether, 1983). Therefore, to describe the motion of particles in a curved spacetime, the metric tensor is defined on a non-Riemanian manifold. The metric tensor is a solution for the Einstein field's equatio which is a second order differential quations on $\mathbb{R}^{4\times4}$.

The metric tensor $g_{\nu\mu}$ (in Einstein index notation) is a 4×4 symmetric tensor that describes the physical length of an infinitesimal line given a particular system of coordinates such that

$$ds^2 = g_{\nu\mu} dx^\nu dx^\mu, \tag{2.1}$$

where $\{dx^\mu\}_{0,1,2,3}$ is a basis for the contangent bundle of the manifold. In classical terms, $\{dx^\mu\}_{0,1,2,3}$ is the differential along a coordinate axis $\{x^\mu\}_{0,1,2,3}$ For a generalized space time, the path of a free particle is described by the geodesic equation. If the coordinates of a particle in spacetime x^μ is parameterized with an affine parameter λ, then the coordinates $x^\mu(\lambda)$ is constrained by an ordinary differential equation, namely the geodesic equation given as:

$$\frac{d^2 x^\mu}{d\lambda^2} + \Gamma^\mu_{\rho\sigma} \frac{dx^\rho}{d\lambda} \frac{dx^\sigma}{d\lambda} = 0, \tag{2.2}$$

where $\Gamma^\mu_{\rho\sigma}$ are the Christoffel symbols given in terms the metric tensor such that

$$\Gamma^\sigma_{\mu\nu} = \frac{1}{2} g^{\sigma\rho} \left(\partial_\mu g_{\nu\rho} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu} \right). \tag{2.3}$$

In the Newtonian limit, the gravitational potential Φ is related to the mass density ρ in the following equation

$$\nabla^2 \Phi = 4\pi G \rho, \tag{2.4}$$

with G being the gravitaional constant. To formulate a theory which reduces to Eq. 2.4, we define the energy-momentum tensor $T_{\mu\nu}$ which is a a 4×4 symmetric tensor that describes the energy density and shear in the four-dimensional spacetime. Since the gravitational field manifest itself as curvature, the analogy to Eq. 2.4 motivate a proportionality relationship between the second derivative of the metric tensor $g_{\mu\nu}$ and the stress-energy tensor $T_{\mu\nu}$

$$\left[\nabla^2 g \right]_{\mu\nu} \propto T_{\mu\nu}. \tag{2.5}$$

A geometric qunatity that is composed of the second derivatives of the matrix tensor is the Ricci tensor $R_{\mu\nu}$ expressed in terms of the Christoffel symbols

$$R_{\mu\nu} = \partial_\lambda \Gamma^\lambda_{\nu\mu} - \partial_\nu \Gamma^\lambda_{\lambda\mu} + \Gamma^\lambda_{\lambda\rho} \Gamma^\rho_{\nu\mu} - \Gamma^\lambda_{\nu\rho} \Gamma^\rho_{\lambda\mu}. \tag{2.6}$$

Meanwhile, the Ricci scale is defined as the contraction of the Ricci tensor

$$R = R^\mu_\mu = g^{\mu\nu} R_{\mu\nu}. \tag{2.7}$$

Therefore, a possible relationship between the stress-energy tensor and the Ricci tensor is

$$R_{\mu\nu} = \kappa T_{\mu\nu}, \tag{2.8}$$

To find κ, the conservation of energy is invoked as

$$\nabla^\mu T_{\mu\nu} = 0, \tag{2.9}$$

where ∇^μ is the covariant derivative. Following Eq. 2.8 and using Eq. 2.6 and Eq. 2.7, the covariant

derivative of the Ricci tensor is

$$\nabla^{\mu} R_{\mu\nu} = \frac{1}{2} \nabla^{\mu} R g_{\mu\nu}. \tag{2.10}$$

Therfore, we can define another tensor $G_{\mu\nu}$ with a vinishing covariant derivative

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}. \tag{2.11}$$

$G_{\mu\nu}$ is called the Einstein tensor, and the Einstein's field equation regardless

$$G_{\mu\nu} = 8\pi G T_{\mu\nu} \tag{2.12}$$

where I used $\kappa = 8\pi G$ which can be obtained by reducing Eq. 2.12 to the Newtonian limit in Eq. 2.4. Einstein modified Eq. 2.12 to account to a vaccum solution by adding a term to represent the vaccum enrgy $\Lambda g_{\mu\nu}$ where Λ is called the cosmological constant, and the final equation becomes

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu} \tag{2.13}$$

Finally, the 4-velocity of a particle in general relativity is defined as

$$U^{\mu} = \frac{dx^{\mu}}{d\lambda}, \tag{2.14}$$

Therfore, the 4-velocity would follow the normalization

$$g_{\mu\nu} U^{\mu} U^{\nu} = -1 \tag{2.15}$$

and the 4-momentum for a particle of mass m is defined as

$$P^{\mu} = m U^{\mu} \tag{2.16}$$

and the normalization of momentum is

$$g_{\mu\nu} P^{\mu} P^{\nu} = -m^2 \tag{2.17}$$

Notice that either of the previous normalization is invariant under coordinate transformation. We define the observed enrgy of a particle of 4-momentum P^{μ} with respect to an observer wiht 4-velocity U^{μ} such that it reduces to the special relativitstic limit if the particle has a vanishing spatial velocity and the observer

$$E_{obs} = -g_{\mu\nu} U^{\mu} P^{\nu}. \tag{2.18}$$

The Einstein theory of general relativity can explain cosmological observation such as cosmological redshift and the expansion of the universe where the expansion is driven by the energy content in the universe. Though the FLRW metric (explained in 2.2) is phenomonologically motivated, the variation of the scale factor with conformal time a is described by the Einstein's field equation.

2.2 The Friedmann-Robertson-Lemaître-Walker metric

To descibe the entire universe using on metric tensor, requires educated simplificaitons and assumptions. The first guiding principle is the Copernican principle (Bondi and Roxburgh, 2010) which states that **'we do not occupy a special place in space'**. The Copernican principle motivates the cosmological principle which states that 'the universe is homogenous and isotropic'. Early galaxy surveys have shown that the universe is only homogeneous on large scales, therefore, the full cosmological principle is stated as **'the universe is homogeneous and symmetric on large scales'**. An additional assumption that simplifies that narrows down the possible formula for a metric is Weyl's postulate which states that **'the world lines of the galaxies form a 3-sheaf of non-intersecting geodesics orthogonal to layers of spacelike hypersurfaces'**. Weyl's posutalete, coupled with the cosmological principle, allows a better definition on where homogeneity and isotropy applies, name that the universe is 'spatially homogeneous and isotropic'. Therefore, a maximally symmetric metric that follows both principles inserts

$$ds^2 = -c^2 dt^2 + g_{ij} dx^i dx^j, \tag{2.19}$$

where t is cosmic time. A two-dimensional schematic diagram of this metric is shown in 2.1.

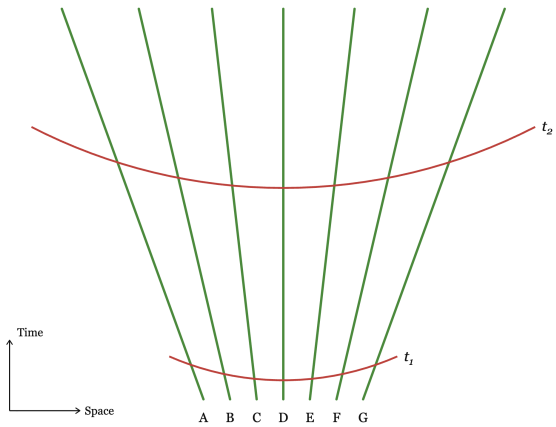

Figure 2.1: A schematic representation of spacetime that follows Weyl's postulate. The green lines labelled with letters are world lines, the path of freely falling objects in the spacetime (or geodesics). The redlines are spacelike hypersurfaces labelled with the same cosmic time t_i. These hypersuraces are homogeneous and isotropic.

Using the tools of differntial geometry, a maximally symmetric metric that also accounts for a homogeneous expansion of the universe is

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + a^2(t)\mathrm{d}s_3^2 \tag{2.20}$$

where $a(t)$ is the scale factor that couples the spatial part with time and the spetial part, $\mathrm{d}s_3^2$ represent a maximally symmetric space that takes the following form in hyperspherical coordinates

$$\mathrm{d}s_3^2 = \mathrm{d}\chi^2 + S_k(\chi)^2 \left(\mathrm{d}\theta^2 + \sin^2\theta\, \mathrm{d}\phi^2\right). \tag{2.21}$$

the function $S_k(\chi)$ takes one of three forms by viture of the number k

$$S_k(\chi) = \begin{cases} \frac{1}{\sqrt{k}} \sin(\sqrt{k}\chi) & \text{if } k > 0, \\ \chi & \text{if } k = 0, \\ \frac{1}{\sqrt{|k|}} \sinh(\sqrt{|k|}\chi) & \text{if } k < 0. \end{cases} \tag{2.22}$$

The case when $k = 0$ represents a flat universe, where the spatial part reduces to the usual spherical coordinates.The coordinate χ could be understood by considering two space like points in spacetime, one with $\chi = 0$ and another with $\chi = \chi_c$ leading To

$$S = a(t)\chi_c, \tag{2.23}$$

which means that the physical distance changes with time for a given χ_c. We call χ_c the comoving distance. Taking the derivative with respect to cosmic time in Eq. 2.23, we can retrieve the empirical law that Hubble (1929) discovered by recording the relationship between the distance to galaxy and the speed of the galaxy

$$\frac{dS}{dt} = V = \frac{\dot{a}}{a} S, \tag{2.24}$$

with $\dot{a} = da/dt$. We define the Hubble parameter at a cosmological time t such that

$$H(t) = \frac{\dot{a}}{a}. \tag{2.25}$$

Another distance indicater is gravitional redshift, similar to doppler redshift, gravitational redshift is the modulation of light frequency as light travels across the universe. If a fudamental oberver at cosmic time t_0 emits a burst of light, then the energy of this light particle is given by Eq. 2.18. If another fundamental observer observes this light at a later time t, the ratio between the two energies is defined as

$$1 + z = \frac{E_e}{E_o}, \tag{2.26}$$

where E_e and E_o are the energy of light measured by the emitter and the observer respectively, and z is called the cosmological redshift. z is related to the wavelength of light in the limit where each of the emeitter and the observer are fundamental observers, meaning that their 4-velocity is $U^\mu = (1, 0, 0, 0)$. Therfore, the light energy could be expressed using Planck's law (Planck, 1900), $E = h/\lambda$, with $h = 6.626 \times 10^{-34}$J·s is Planck's constant, and λ is the wavelength of light. The final relationship describing the modulation in the observed wavelength of light is

$$1 + z = \frac{a(t_o)}{a(t_e)} = \frac{\lambda_o}{\lambda_e} \tag{2.27}$$

with λ_e and λ_o are the light wavelength measured by the emitter and the observer respectively. This relationship could be exploited by quantifying the modulation of the wavelength and its relationship wo the scale factor. In a nutshell, a light emitted at a scale factor $a(t_e)$ travels through an expanding universe until measured by another observer at time $t_o > t_e$ correponding to $a(t_o) > a(t_e)$, therfore $\lambda_o > \lambda_e$, which explains why distant sources are redshifted. In practive, by measuring the sptrum of light of a distant galaxy and convolving it with a standard spectrum (the expected spectrum of galaxy

in the nearby universe), it it possible to obtain at the redshift of the distant galaxy. If we choose $a = 1$ in the current epoch, the relationship between redshift and the scale factor reduces to

$$a = \frac{1}{1+z}, \tag{2.28}$$

where $z = 0$ refers to the current epoch. Since light move along a null geodesic, is possible to obtain the comocving distance to an emitter given the redshift that the emitter. Starting from Eq. 2.20 and Eq. 2.21, and working in a coordinate system with the light travelling along $\theta = 0$ and $\phi = 0$, the null geodesic dictates that

$$cdt = a(t)d\chi \tag{2.29}$$

which could be further aranged using Eq. 2.25 and Eq. 2.28 into

$$\chi = \int_1^{z_e} \frac{c}{(1+z)H(z)} dz \tag{2.30}$$

Throughout this chapter, a question is lurking behind all the kinemetics of the universe that the FLRW metric proposes: how does the scale factor a changes with cosmic time? In principle, this requires using the Einstein's field equation Eq. 2.13 along with assumptions about the form of the stress energy tensor $T^{\mu\nu}$. Assuming that the energgy density in the universe is that of a perfect fluid, solving the Einstein's field equation for a yields

$$\begin{aligned} \frac{\ddot{a}}{a} &= -\frac{4\pi G}{3}\left(\rho + \frac{3p}{c^2}\right) + \frac{\Lambda c^2}{3}, \\ \frac{\dot{a}^2}{a^2} &= \frac{8\pi G}{3}\rho + \frac{\Lambda c^2}{3} - \frac{kc^2}{a^2}. \end{aligned}$$

with p being the isotrpic pressure of the cosmic fluid and ρ being the energy density. These equations are known as teh Friedmann equations, describing the evolution of the scale factor a with respect to cosmic time t. To solve these equations, a relationship between the pressure p and the energy density is needed, known as the equation of state. The possible components that contribute to the energy density ρ have been a process of trial and error starting de Sitter universe (Sitter, 1917) which is a universe with no matter up to the recently speculated model with dynamical dark energy (Adame et al., 2025). In this thesis, we consider the ΛCDM cosmoligcal model and describe its components in 2.3.

2.3 The cosmoligcal model

- ΛCDM, each component and its equation of state.
- Planck results.
- The Einstein Boltzmann equation and linear theory.
- cosmological tensions
- The cosmological fluid the Einstein-Boltzmann Equation (fluid equation, the linear pertubation theory, the non-linear theory)

- matter power spectrum
- angular power spectra (introduce the limber and the non-limber integrals)

2.4 Weak gravitational Lensing

2.5 The impact of baryonic feedback on the large-scale structure

results from hydrodynamical simulations, cosmology analyses resutls. maybe cite Fabian's papeer

2.6 Fast radio bursts

describe how modelling the PDF is a hard thing to do To forward model observations, we convolve the $DM - z$ represented by $P(DM|z)$ with measurement and systematic uncertainties, in addition to other observational effects. A catalog of detected FRBs is expected to have observed dispersion measure $DM^i_{\rm obs}$, galactic coordinates, and redshifts for a subset of and observations z^i. We assume that each detected FRB was emitted at redshift z^i where the difference between identified and unidentified host can be integrated in how $P(z^i)$ is modeled. Assume that we observe N_{FRB} fast radio bursts signal. We express the observations as the probability of detecting an FRB $DM_{\rm obs}$ given a set of estimated redshifts $\{z^i\}$, can be decomposed using the Bayes rule such that

$$P(DM_{\rm obs}|\{z^i\}) \propto P(\{z^i\}|DM_{\rm obs})P(DM_{\rm obs}) \tag{2.31}$$

Furthermore, if we assume that each observation z_i is independent of other observations we may use the Bayes rule to decompose $P(\{z^i\}|DM_{\rm obs})$ such that

$$P(\{z^i\}|DM_{\rm obs}) = \prod_i P(z^i|DM_{\rm obs}) \tag{2.32}$$

Furthermore, the individual terms in Eq. 2.32 could be derived using the joint probability distribution of the observed and the dispersion measure contribution from the large scale structure, namely $DM_{\rm obs}$ and DM such that

$$\begin{aligned} P(z^i|DM_{\rm obs}) &\propto P(DM_{\rm obs}|z^i)P(z^i) \\ &= P(z^i)\int P(DM_{\rm obs}, DM|z^i)dDM \\ &= P(z^i)\int P(DM_{\rm obs}|z^i, DM)P(DM|z^i)dDM \end{aligned} \tag{2.33}$$

The final expression that relates the probability distribution function of an observation to other modelled terms is

$$\begin{aligned} &P(DM_{\rm obs}|\{z^i\}) \propto \\ &\qquad P(DM_{\rm obs})\prod_i P(z^i)\int P(DM_{\rm obs}|z^i, DM)P(DM|z^i)dDM \end{aligned} \tag{2.34}$$

Furthermore, to model each component of the resulting expression, we start with $P(DM_{\text{obs}})$. There are three components that contribute additively to DM_{obs}: the large scale structure contribution DM_{LSS}, the Milkyway contribution DM_{MW}, and the host galaxy contribution DM_{host}, and the observed dispersion measure is given as

$$DM_{\text{obs}} = DM_{\text{LSS}} + DM_{\text{MW}} + DM_{\text{host}} \tag{2.35}$$

For simplicity, we indicate to the contribution from the large-scale structure simply with DM which is the same in Equation Eq. 2.34. Consider the joint probability distribution $P(DM, DM_{\text{MW}}, DM_{\text{host}})$, this can be related to $P(DM_{\text{obs}}) = P(DM + DM_{\text{MW}} + DM_{\text{host}})$ via the constrained integral

$$\begin{aligned} P(DM_{\text{obs}}) &= P(DM + DM_{\text{MW}} + DM_{\text{host}}) \\ &= \int_C P(DM, DM_{\text{MW}}, DM_{\text{host}})\, \mathrm{d}DM\, \mathrm{d}DM_{\text{MW}}\, \mathrm{d}DM_{\text{obs}} \\ &= \int_C P(DM)P(DM_{\text{MW}})P(DM_{\text{host}})\, \mathrm{d}DM\, \mathrm{d}DM_{\text{MW}}\, \mathrm{d}DM_{\text{obs}} \end{aligned} \tag{2.36}$$

Where $C = \{DM, DM_{\text{MW}}, DM_{\text{host}}; DM_{\text{obs}} = DM + DM_{\text{MW}} + DM_{\text{host}}\}$ Similarly, I can model $P(DM_{\text{obs}}|z^i, DM)$ as

$$\begin{aligned} P(DM_{\text{obs}}|z^i, DM) = \int_C & P(DM|z^i, DM)P(DM_{\text{MW}}|z^i, DM) \\ & P(DM_{\text{host}}|z^i, DM)\, \mathrm{d}DM\, \mathrm{d}DM_{\text{MW}}\, \mathrm{d}DM_{\text{obs}} \end{aligned} \tag{2.37}$$

Each of the terms making up the integral could be further reduced

$$P(DM|z^i, DM) \propto P(DM|z^i)P(DM) \tag{2.38}$$

$$P(DM_{\text{MW}}|z^i, DM) = P(DM_{\text{MW}}) \tag{2.39}$$

$$P(DM_{\text{host}}|z^i, DM) \propto P(DM_{\text{host}}|z^i)P(DM|z^i) \tag{2.40}$$

Keeping in mind, that DM_{host} that we use in Eq. 2.34 is flat since we do not have any information of the

CHAPTER 3

Methodology

In this section, we explain the methoddology used in this thesis.

Add a flow chart that summarizes the different steps in the pipeline, like the one in the interview presentation.

Consider adding a table which describes the fiducial cosmology.

3.1 Modelling the electron power spectrum

3.1.1 The baryonification correction model

explain how we ignore the redshift evolution of some parameters.

3.1.2 The halo model

Check that you have explained what dark matter halos are in the beginning. Refer to Either the review by Marika or the paper on the Web halo model.

The Core Cosmology Library (CCL)

3.2 Simulating the dispersion measure signal

Add a table with the fiducial values and their prior. explain how we ignore the redshift evolution of some parameters.

3.2.1 the Generator for Large Scale Structure (GLASS)

3.2.2 The electron angular power spectra

3.2.3 The cosmological covariance of the dispersion measure signal

3.3 The systematic uncertainties

It is important here to mention how this is modelled on the level of the simulation and the level of covariance.

3.3.1 Redshift uncertainty

3.3.2 Selection effects

3.3.3 Unlocalised hosts

3.3.4 Milkyway contribution

3.3.5 Host galaxy contribution

3.3.6 The lensing of the Dispersion measure signal

3.4 Simulation-based inference

3.4.1 Approximate Baysian inference

3.4.2 Neural Posterior Estimator

3.4.3 The Sequentual Neural Posterio Estimator

The truncated sequential neural posterior estimator

3.4.4 Flow matching posterior estimator

3.4.5 Data Compression

I simulate the electron density shells and place FRBs in the same redshift and angular position as in the mock catalog. This will reduce the computational cost and the need to calculate the covariance matrix for each realisation within the SBI pipeline. (Try to do this excercise without choosing the sdame redshifts for a lower number of sources such that it lowers down the computational costs)

Numerical derivative stability

The Sensitivity of compressed statistics to model parameters

Add a plot about derivative stability. Add a plot for the sensitivity.

CHAPTER 4

Results

CHAPTER 5

Discussion & Conclusion

APPENDIX A

Useful information

In the appendix you usually include extra information that should be documented in your thesis, but not interrupt the flow.

The LaTeX WikiBook (**latexwiki**) is a useful source of information on LaTeX.

Bibliography

Adame, A. et al. (2025), *DESI 2024 VI: cosmological constraints from the measurements of baryon acoustic oscillations*, Journal of Cosmology and Astroparticle Physics **2025** 021 (cit. on p. 12).

Alonso, D. (2021), *Linear anisotropies in dispersion-measure-based cosmological observables*, Physical Review D **103** (cit. on p. 6).

Bahcall, J. N. and L. Spitzer (1982), *The space telescope*, Scientific American **247** 40 (cit. on p. 4).

Bartelmann, M. and P. Schneider (2001), *Weak gravitational lensing*, Physics Reports **340** 291 (cit. on p. 4).

Bennett, C. L. et al. (2003), *First-year Wilkinson Microwave Anisotropy Probe (WMAP) observations: Foreground emission*, The Astrophysical Journal Supplement Series **148** 97 (cit. on p. 4).

Boggess, N. W. et al. (1992), *The COBE mission - its design and performance two years after launch*, The Astrophysical Journal **397** 420 (cit. on p. 4).

Bondi, H. and I. W. Roxburgh (2010), *Cosmology*, Courier Corporation (cit. on p. 10).

Carroll, S. M. (2024), *Spacetime and geometry: An introduction to general relativity*, Cambridge University Press (cit. on p. 7).

Daalen, M. P. van, J. Schaye, C. Booth and C. Dalla Vecchia (2011), *The effects of galaxy formation on the matter power spectrum: a challenge for precision cosmology*, Monthly Notices of the Royal Astronomical Society **415** 3649 (cit. on pp. 4, 5).

Davis, M., J. Huchra, D. W. Latham and J. Tonry (1982), *A survey of Galaxy redshifts. II - the large scale space distribution*, The Astrophysical Journal **253** 423 (cit. on p. 3).

Davis, M. and P. J. Peebles (1983), *A survey of Galaxy redshifts. V - the two-point position and velocity correlations*, The Astrophysical Journal **267** 465 (cit. on p. 3).

Di Valentino, E. et al. (2025), *The CosmoVerse White Paper: Addressing observational tensions in cosmology with systematics and fundamental physics*, Physics of the Dark Universe **49** 101965 (cit. on p. 3).

Dodelson, S. and F. Schmidt (2026), *Modern cosmology*, Academic Press (cit. on pp. 4, 7).

Eifler, T. et al. (2015), *Accounting for baryonic effects in cosmic shear tomography: Determining a minimal set of nuisance parameters using PCA*, Monthly Notices of the Royal Astronomical Society **454** 2451 (cit. on p. 5).

Hu, W. and M. Tegmark (1999), *Weak lensing: Prospects for measuring cosmological parameters*, The Astrophysical Journal **514** (cit. on p. 4).

Hubble, E. (1929), *A relation between distance and radial velocity among extra-galactic nebulae*, Proceedings of the National Academy of Sciences **15** 168 (cit. on pp. 3, 11).

Jain, B. and A. Taylor (2003), *Cross-correlation tomography: Measuring dark energy evolution with weak lensing*, Physical Review Letters **91** (cit. on p. 4).

Kaiser, N. and G. Squires (1993), *Mapping the dark matter with weak gravitational lensing*, The Astrophysical Journal **404** 441 (cit. on p. 4).

Konar, K., R. Reischke, S. Hagstotz, A. Nicola and H. Hildebrandt (2025), *Constraining the dispersion measure redshift relation with simulation-based inference*, The Open Journal of Astrophysics **8** (cit. on p. 6).

Konietzka, R. M. et al. (2025), *Ray-tracing Fast Radio Bursts Through IllustrisTNG: Cosmological Dispersion Measures from Redshift 0 to 5.5*, arXiv: `2507.07090 [astro-ph.CO]` (cit. on p. 6).

Kovač, M. et al. (2025), *Baryonification II: Constraining feedback with X-ray and kinematic Sunyaev-Zel'dovich observations*, Journal of Cosmology and Astroparticle Physics **2025** 046 (cit. on p. 5).

Krause, E. et al. (2021), *Dark energy survey year 3 results: Multi-probe modeling strategy and validation*, arXiv preprint arXiv:2105.13548 (cit. on p. 5).

Lorimer, D. R., M. Bailes, M. A. McLaughlin, D. J. Narkevic and F. Crawford (2007), *A bright millisecond radio burst of extragalactic origin*, Science **318** 777 (cit. on p. 5).

Luke Hoffmann, J. et al. (2024), *Modelling DSA, fast, and craft surveys in a z-dm analysis and constraining a minimum FRB Energy*, Publications of the Astronomical Society of Australia **42** (cit. on p. 6).

Maoli, R. et al. (2001), *Cosmic Shear Analysis in 50 uncorrelated VLT Fields. implications for ω_0, σ_8*, Astronomy and Astrophysics **368** 766 (cit. on p. 4).

Maraio, A., A. Hall and A. Taylor (2025), *Mitigating baryon feedback bias in cosmic shear through a theoretical error covariance in the matter power spectrum*, Monthly Notices of the Royal Astronomical Society **537** 1749 (cit. on p. 5).

Moreira, M. G. et al. (2021), *Mitigating baryonic effects with a theoretical error covariance*, Monthly Notices of the Royal Astronomical Society **507** 5592 (cit. on p. 5).

Noether, E. (1983), *Invariante Variationsprobleme*, Springer Collected Works in Mathematics 231 (cit. on p. 7).

Petroff, E., J. W. Hessels and D. R. Lorimer (2019), *Fast radio bursts*, The Astronomy and Astrophysics Review **27** (cit. on p. 5).

Piccirilli, G., M. Zennaro, C. García-García and D. Alonso (2025), *Robust cosmic shear with small-scale nulling*, Journal of Cosmology and Astroparticle Physics **2025** 017 (cit. on p. 5).

Planck, M. (1900), *Zur Theorie des Gesetzes der Energieverteilung im Normalspektrum*, Verhandlungen der Deutschen Physikalischen Gesellschaft **2** 237 (cit. on p. 11).

Pope, A. (2004), *Cosmological parameters from Eigenmode Analysis of Sloan Digital Sky Survey Galaxy Redshifts*, AIP Conference Proceedings **743** 120 (cit. on p. 4).

Reischke, R. and S. Hagstotz (2025), *A first measurement of baryonic feedback with Fast Radio Bursts*, arXiv: `2507.17742 [astro-ph.CO]` (cit. on p. 6).

Reischke, R., D. Neumann, K. A. Bertmann, S. Hagstotz and H. Hildebrandt (2023), *Calibrating baryonic feedback with weak lensing and fast radio bursts*, arXiv preprint arXiv:2309.09766 (cit. on p. 6).

Riess, A. G. et al. (1998), *Observational evidence from supernovae for an accelerating universe and a cosmological constant*, The Astronomical Journal **116** 1009 (cit. on p. 4).

Rudd, D. H., A. R. Zentner and A. V. Kravtsov (2008), *Effects of baryons and dissipation on the matter power spectrum*, The Astrophysical Journal **672** 19 (cit. on p. 4).

Schneider, A., M. Kovač et al. (2025), *Baryonification: An alternative to hydrodynamical simulations for cosmological studies*, Journal of Cosmology and Astroparticle Physics **2025** 043 (cit. on p. 5).

Schneider, A. and R. Teyssier (2015), *A new method to quantify the effects of baryons on the matter power spectrum*, Journal of Cosmology and Astroparticle Physics **2015** 049 (cit. on p. 5).
Schneider, P., T. Eifler and E. Krause (2010), *COSEBIs: Extracting the full E-/B-mode information from cosmic shear correlation functions*, Astronomy and Astrophysics **520** (cit. on p. 4).
Schneider, P., J. Ehlers and E. E. Falco (1992), *Gravitational lenses*, Springer International Publishing (cit. on p. 4).
Schneider, P., L. Van Waerbeke, B. Jain and G. Kruse (1998), *A new measure for Cosmic Shear*, Monthly Notices of the Royal Astronomical Society **296** 873 (cit. on p. 4).
Seitz, C. and P. Schneider (1996), *Steps towards nonlinear cluster inversion through gravitational distortions: III. Including a redshift distribution of the sources*, arXiv preprint astro-ph/9601079 (cit. on p. 4).
Sharma, K. et al. (2026), *Probing baryonic feedback and cosmology with the 3 × 2-point statistic of fast radio bursts and galaxies*, The Astrophysical Journal **998** 109 (cit. on p. 6).
Sitter, W. de (1917), *On Einstein's theory of gravitation and its astronomical consequences. third paper.*, Monthly Notices of the Royal Astronomical Society **78** 3 (cit. on p. 12).
Smoot, G. F. (1999), *COBE observations and results*, Conference on 3K cosmology 1 (cit. on p. 4).
Takahashi, R. (2024), *Impact of lensing bias on the cosmological dispersion measure*, Journal of Cosmology and Astroparticle Physics **2024** 057 (cit. on p. 6).
Torkamani, M. et al. (2026), *Baryonification III: An accurate analytical model for the dispersion measure probability density function of fast radio bursts*, arXiv preprint arXiv:2601.18784 (cit. on p. 6).
Totsuji, H. and T. Kihara (1969), *The correlation function for the distribution of galaxies*, Publications of the Astronomical Society of Japan **21** 221 (cit. on p. 3).
Turner, M. S. (2001), *Dark matter and dark energy in the universe*, Particle Physics and the Universe 210 (cit. on p. 4).

List of Figures

2.1 A schematic representation of spacetime that follows Weyl's postulate. The green lines labelled with letters are world lines, the path of freely falling objects in the spacetime (or geodesics). The redlines are spacelike hypersurfaces labelled with the same cosmic time t_i. These hypersuraces are homogeneous and isotropic. 10

List of Tables

Acknowledgements

I am strongly indebted to my supervisors Dr. Andrina Nicola and Dr. Robert Reischke for their immense support throughout the duration my master's studies. Their support and advice were pivotal in the process of conducting this work. I would also like to thank the member of the 'Combining Cosmological Probes' group at the Argelander Institute for Astronomy, especially, Subarna, Mokka, Micheal, Reihana, and Reiko.

Furthermore, I would like to thank my friends and colleagues at the University of Bonn who made the master's studies journey more fun and less stressful. I would like to thank my aunt, Sana, who kept my company and offered immense support when adjusting to the life in Germany and Bonn.

I thank my girlfriend, Fée, who was my best friend throughout the journey of my master's thesis. Her words of encourgments were crucial in the moments of doubt.

I would further like to thank the memebers of the physical cosmology department at the Max Planck Institute for Astrophysics where part of this work was conducted, in particular, Fabian Schmidt, Šafak Celik, Lena Scheuchel, Sveva Castillo, Florian List, and Adr

You should probably use `\chapter*` for acknowledgements at the beginning of a thesis and `\chapter` for the end.